

Software Engineering Lab 1
Requirements Engineering & UML Use-Case Modelling

Smart Lab Equipment & Slot Reservation Portal

Problem Statement #01 — Campus & Academic Operations

Name	Abhinav Agraharam
SRN	PES1UG24CS001
Section	5A
Actors modelled	Student, Lab Technician
Deliverables included	Requirements Table (FR-001–005, NFR-001–002) UML Use-Case Diagram Use-Case Flow Specification — Reserve Equipment Slot

1. Requirements Table

Actors: Student, Lab Technician

Functional Requirements

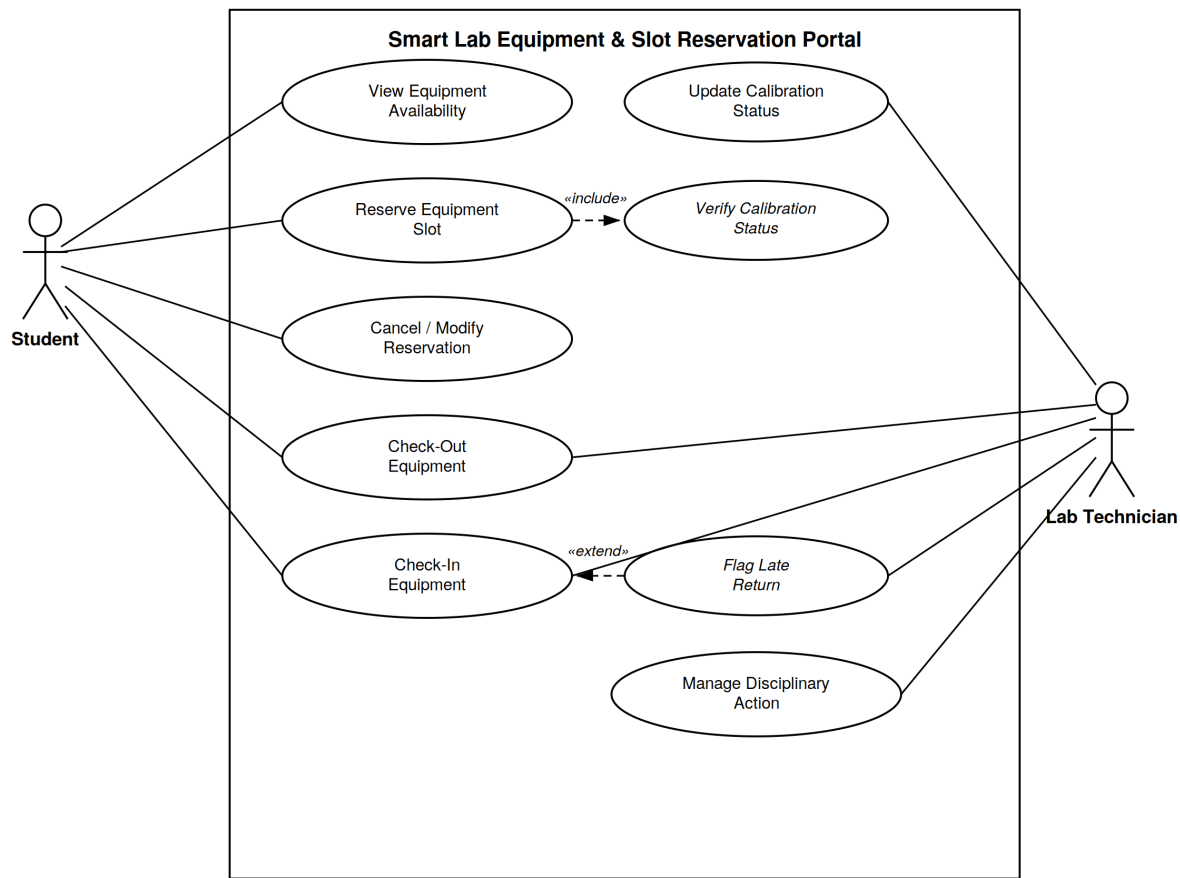
ID	Priority	Description	Acceptance Criteria	Rationale
FR-001	High	Allow authenticated students to view real-time availability and reserve a maximum 2-hour slot for any calibrated equipment up to 7 days in advance. Reject reservations against equipment marked Out of Service or Due for Calibration at submission time.	Pass: slot locked, token generated, calibration check satisfied. Fail: double-booking allowed, slot exceeds 2 hours, or reservation succeeds on uncalibrated equipment.	Prevents scheduling conflicts and ensures students are never assigned instruments unfit for accurate measurements.
FR-002	High	Allow lab technicians to set equipment calibration status to Calibrated, Due for Calibration, or Out of Service, and immediately block new reservations against an item marked Out of Service.	Pass: status change takes effect immediately; existing future reservations on a newly Out-of-Service item are flagged for review. Fail: a student can reserve Out-of-Service equipment.	Keeps availability data trustworthy and prevents students being handed faulty instruments.
FR-003	Medium	Allow a student to cancel or reschedule a reservation up to 2 hours before slot start without penalty; requests after that cutoff still free the slot but are logged as a late cancellation.	Pass: pre-cutoff cancellation frees the slot with no penalty; post-cutoff frees the slot and records a late-cancellation entry. Fail: slot stays locked, or a late cancellation goes unrecorded.	Gives students flexibility while discouraging last-minute cancellations that waste booked lab capacity.
FR-004	High	Record actual equipment return time against the reserved end time, automatically flag any return more than 15 minutes late, and suspend a student's reservation privileges for 7 days after their third late-return flag in a semester.	Pass: late returns are flagged and suspension triggers correctly on the third infraction. Fail: a late return is unflagged, or a suspended student can still book.	Enforces the late-return disciplinary policy named in the problem statement and protects equipment availability for the cohort.
FR-005	Medium	Generate a unique reservation token/QR code at booking confirmation. A lab technician scans or enters it to release equipment at check-out and again to close the reservation at check-in.	Pass: check-out only succeeds against a valid, unexpired token tied to an active reservation; check-in closes that reservation and timestamps the return. Fail: check-out succeeds without a valid token, or check-in succeeds without a prior check-out.	Creates an auditable link between a reservation and physical possession of equipment; FR-004 depends on this timestamp.

Non-Functional Requirements

ID	Type	Description	Acceptance Criteria	Rationale
NFR-001	Performance & Concurrency	Process concurrent slot-reservation requests for the same equipment item using database-level lock isolation (row-level locking or optimistic concurrency control) so no two overlapping reservations can both commit, with 95th-percentile response time under 300 ms at 200 simulated concurrent booking attempts.	Pass: 10,000 simulated concurrent attempts produce zero double-bookings and p95 latency under 300 ms. Fail: any double-booking occurs, or p95 latency exceeds target.	Booking is the system's core transaction; a race condition here reproduces exactly the double-booking failure the portal exists to prevent.
NFR-002	Auditability & Data Integrity	Write an immutable audit-log entry for every reservation, calibration-status change, check-out, check-in, and disciplinary flag; retain each entry at least one academic year; prevent any role, including administrators, from editing or deleting historical entries through the application.	Pass: an audit entry exists for 100% of listed event types; edit/delete attempts on existing entries are rejected. Fail: any listed event fails to log, or an entry can be altered.	FR-004's disciplinary enforcement and any calibration dispute both depend on a record that can't be quietly edited after the fact.

2. UML Use-Case Diagram

Includes at least one «include» relationship (Reserve Equipment Slot → Verify Calibration Status) and one «extend» relationship (Flag Late Return → Check-In Equipment).



3. Use-Case Flow Specification: Reserve Equipment Slot

Actor: Student

Related requirements: FR-001, FR-005; includes the concurrency-safe locking behaviour described by NFR-001 at the commit step.

Preconditions

The student is authenticated on the portal. The equipment catalog and its real-time availability calendar are up to date. The equipment item the student intends to select is recorded in the system with some calibration status (Calibrated, Due for Calibration, or Out of Service) — the reservation itself is what tests that status, so no particular status is assumed going in.

Postconditions

On success: a new reservation record exists for the selected equipment item, date, and time slot; that slot is locked and no longer shown as available to other students; a unique reservation token has been generated and displayed to the student for later use at check-out.

On failure: no reservation record is created, the slot remains (or returns to) available, and the student has been shown the specific reason the request did not go through.

Main Success Scenario

1. The student logs into the portal and opens the equipment catalog.
2. The student selects an equipment category or a specific instrument and views its real-time availability calendar for the next 7 days.
3. The student selects an open slot of up to 2 hours for that item.
4. The system executes the included use case **Verify Calibration Status** against the selected item, which returns Calibrated.
5. The system checks the requested slot against existing reservations for that item to confirm there is no overlap, applying database-level lock isolation so a simultaneous request from another student cannot slip through (per NFR-001).
6. The system locks the slot, creates the reservation record, and generates a unique reservation token.
7. The system displays a confirmation screen with the reservation details and token to the student.

Alternate Flow A1 — Equipment Not Calibrated

Trigger: at step 4, the included use case Verify Calibration Status returns Due for Calibration or Out of Service instead of Calibrated.

1. The system rejects the reservation request before any slot is locked.
2. The system displays the specific reason (equipment awaiting calibration, or out of service) to the student rather than a generic error.
3. If another item of the same equipment type is currently calibrated and has an open slot in the requested window, the system suggests it to the student.
4. The student may select the suggested alternative, in which case the flow resumes at step 3 of the main success scenario, or may cancel, in which case the use case ends with no reservation created.

Note on scope: only one alternate flow is documented here per the submission requirement. A second one worth flagging for completeness — if broader coverage is expected — is a slot-conflict path at step 5 (another student's reservation commits first), mechanically similar to A1 but triggered by the lock check instead of the calibration check.